

Contemporary C++: *Learning Modern C++ in a Modern Way*



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Agenda 14/30

Session 5. Introduction to contiguously sequential data structures: C-Style arrays, C++ vectors, C++ standard arrays:

Part 3

- The C++11 arrays
- Compare C-style arrays with C++ vectors and C++ arrays
- Introduction to C++ arrays: The array containers
- More examples on contiguously sequential data structures
- References and further readings
- Q & A

120 min (incl. Q & A)



Array

- An array, is a fixed-size sequence of elements of a given type where the number of elements is specified at compile-time.
- Like a built-in array, an array is simply a sequence of elements, with no handle:



- This implies that a local array does not use any free store.

- `<array>`
- Compile-time array size

```
template <class T, size_t N> struct array;  
array<int, 3> a;
```

```
array<int, 3> a;  
array<string, 3> language = { "C", "C++", "Python" };  
array<double> d; // error: size not specified
```

GD
Array Accumulate
Prog.

- The standard array is suitable for embedded systems programming and similar constrained, performance-critical, or safety critical tasks.
- (Hard) Real-time systems, Airplanes, ...

Range-based for loop

- Conventional general *for* statement: You can do almost anything.

```
for (initialization-statement conditionopt; expressionopt)  
    statement
```

```
int a[10];  
for (int i = 0; i <=10; ++i) {  
    a[i] = 0;  
}
```

- off-by one error
- Common idiom in C++ programming: Iterator through a container from begin to end
- Rather verbose
- A range-for statement allows you to iterate through a "range".
- Range: arrays, containers, initializer lists, and all data structures with the STL sense.
- C++11 → One step further: range-based for loop

Range-based for loop

1.

```
void print(vector<int>& L)
{
    for(int x : L) cout << x << '\n';
}
```

Expands
to

2.

```
void print(vector<int>& L)
{
    for(auto x : L) cout << x << '\n';
}
```

Iterate through the sequence using *begin* and *end* to find the sequence's bounds.

■ Arrays

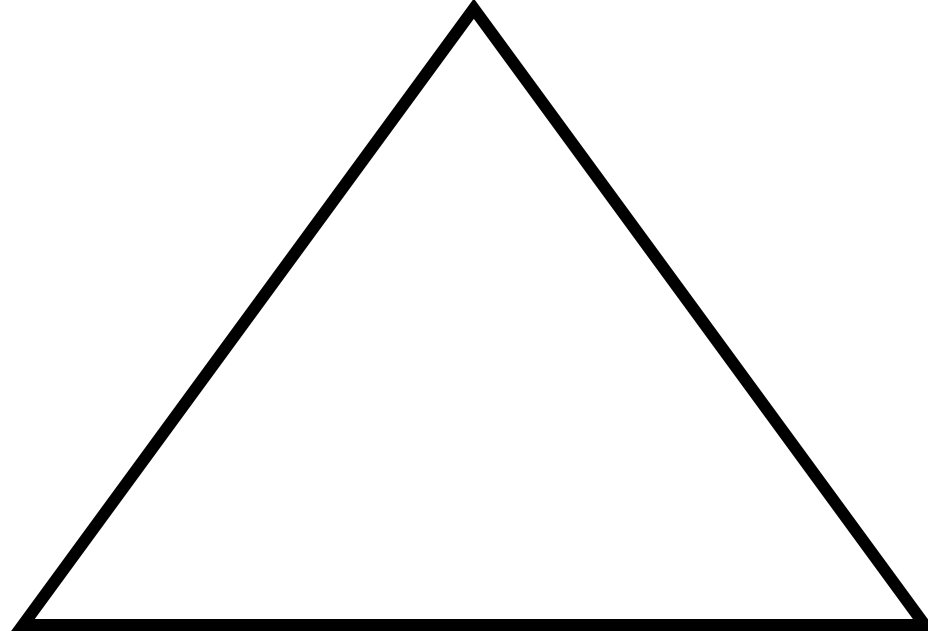
- No off-by-one error

```
int a[10];
int i = 0;
for (auto x : a) x = i++;
for (const auto x : a) cout << x << '\n';
```

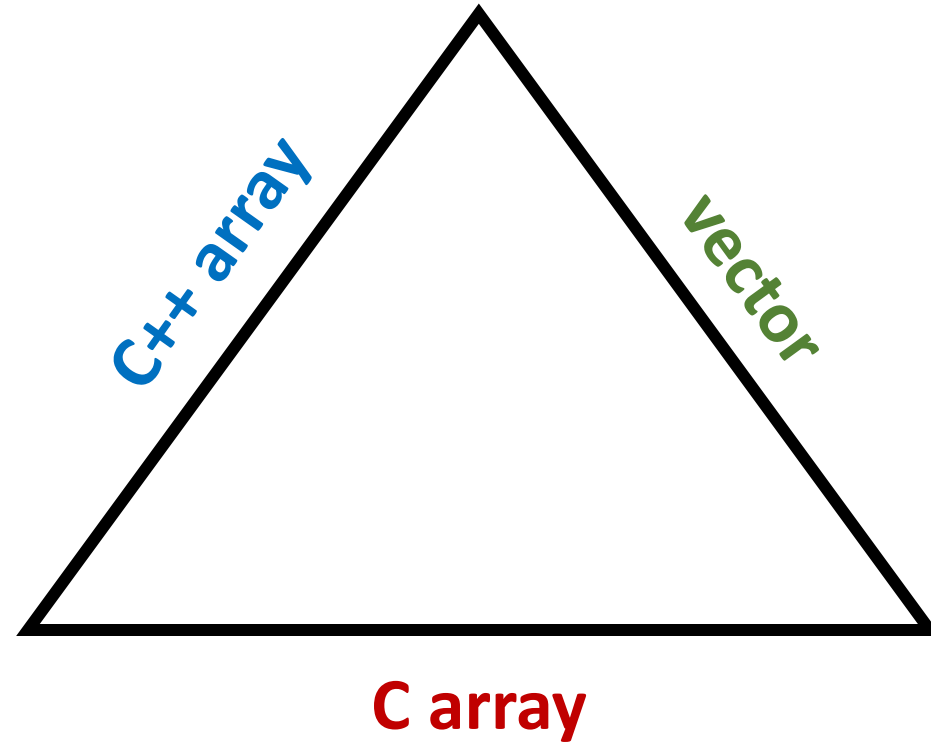
Range-based for loop: more examples

Built-in array vs. Standard array vs. vector

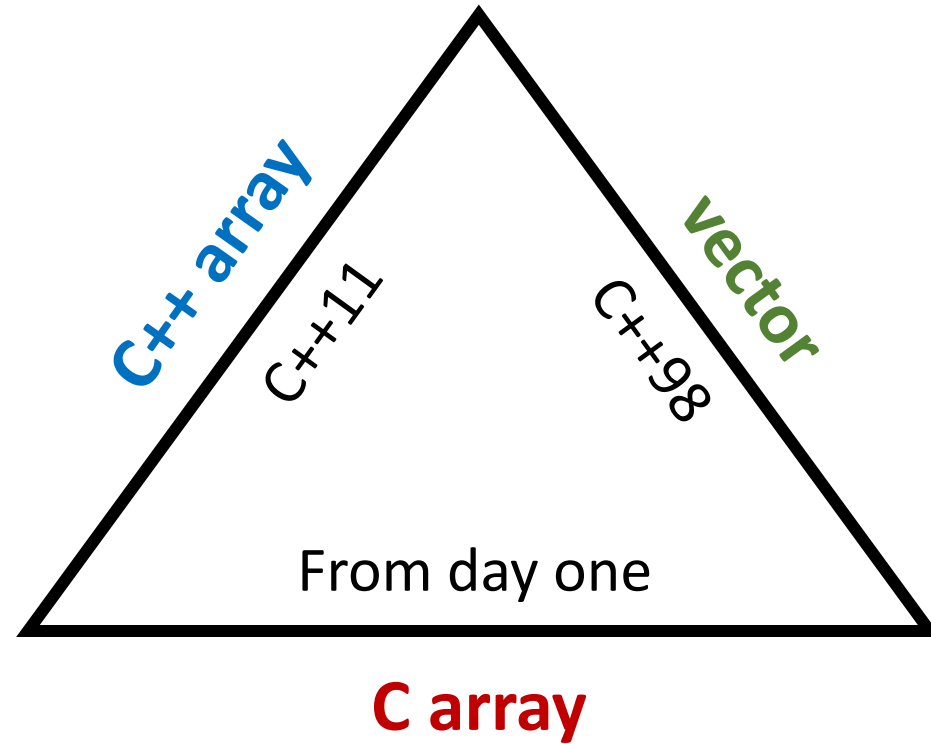
Built-in array vs. Standard array vs. vector



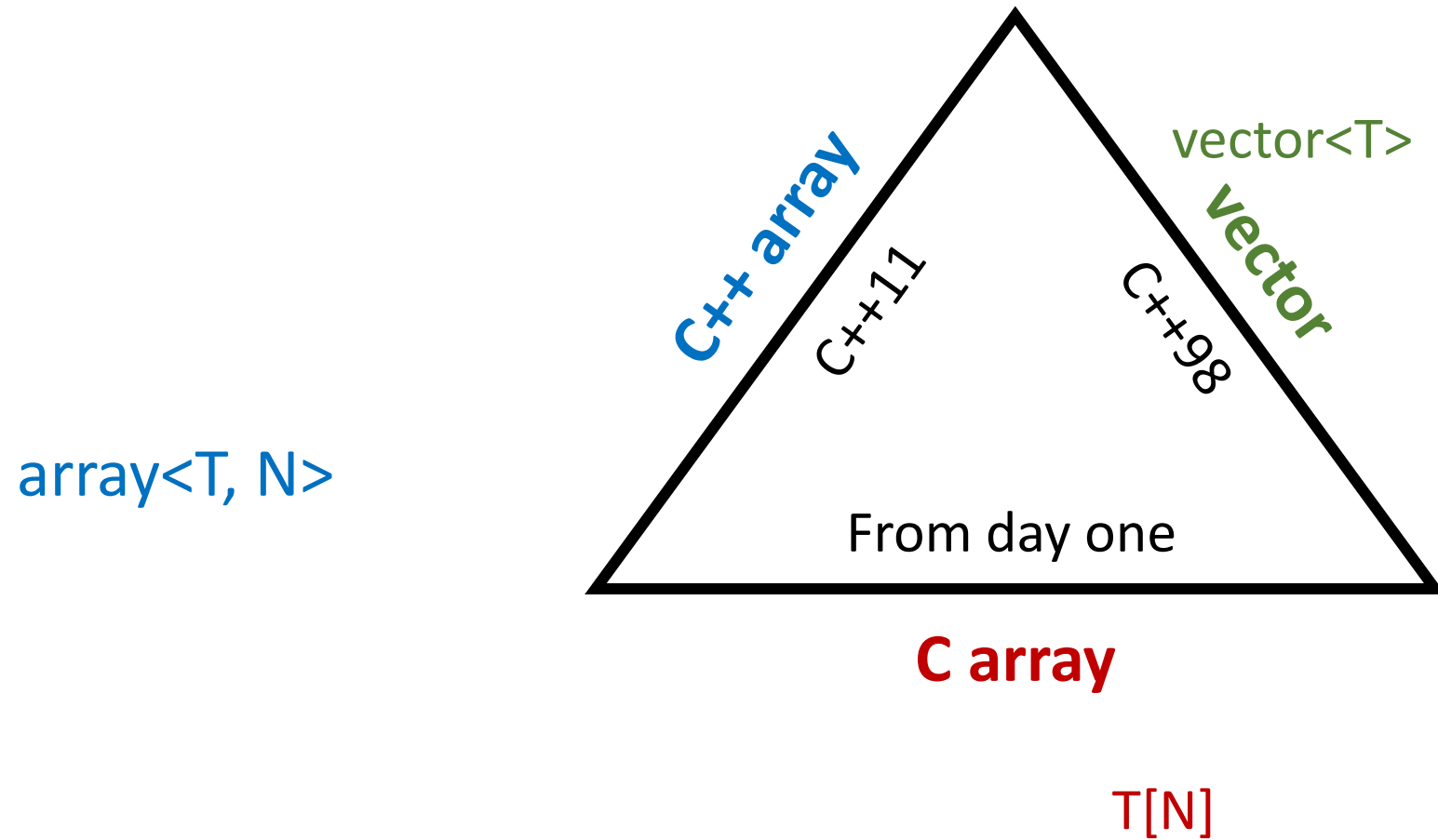
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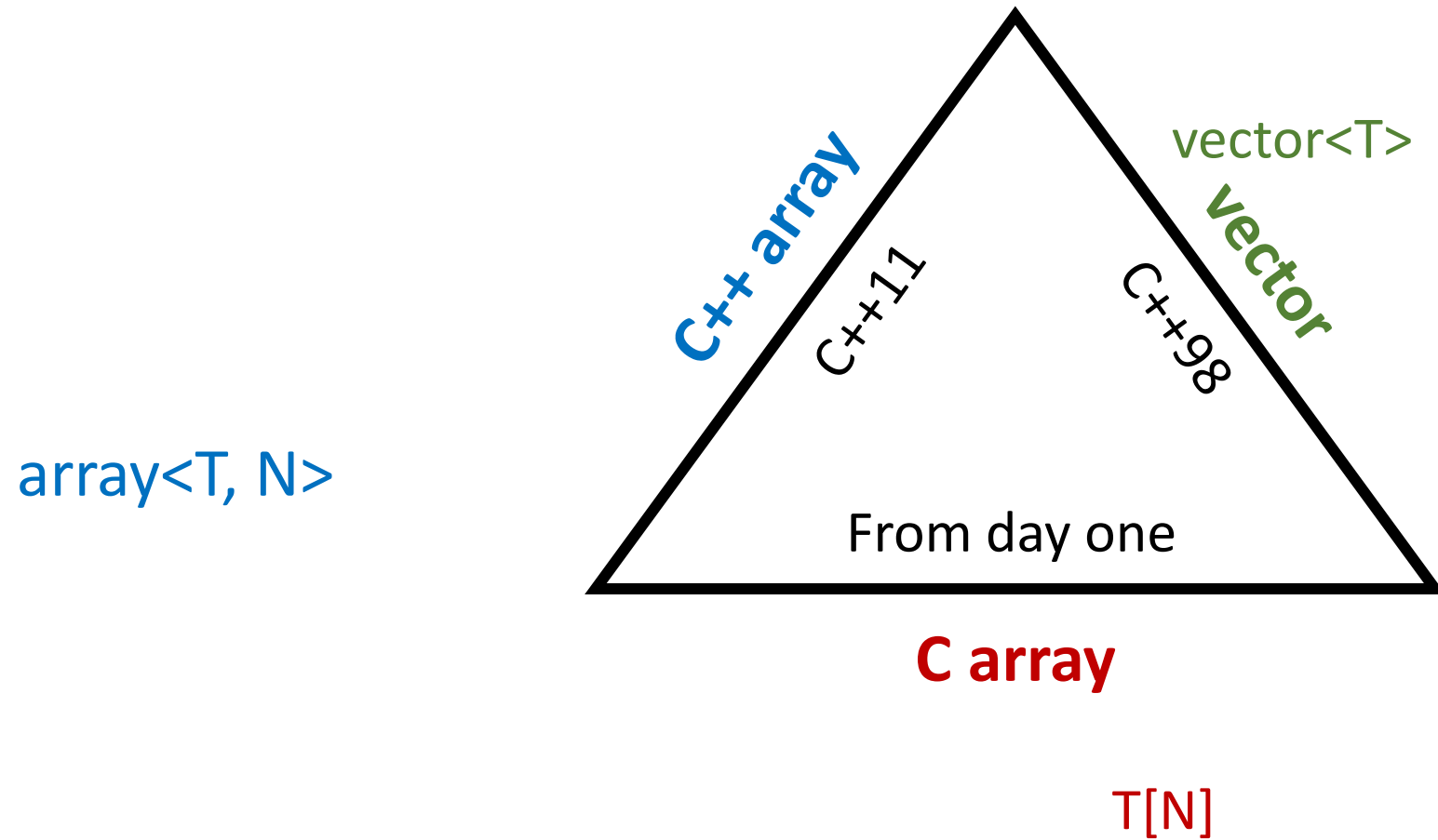
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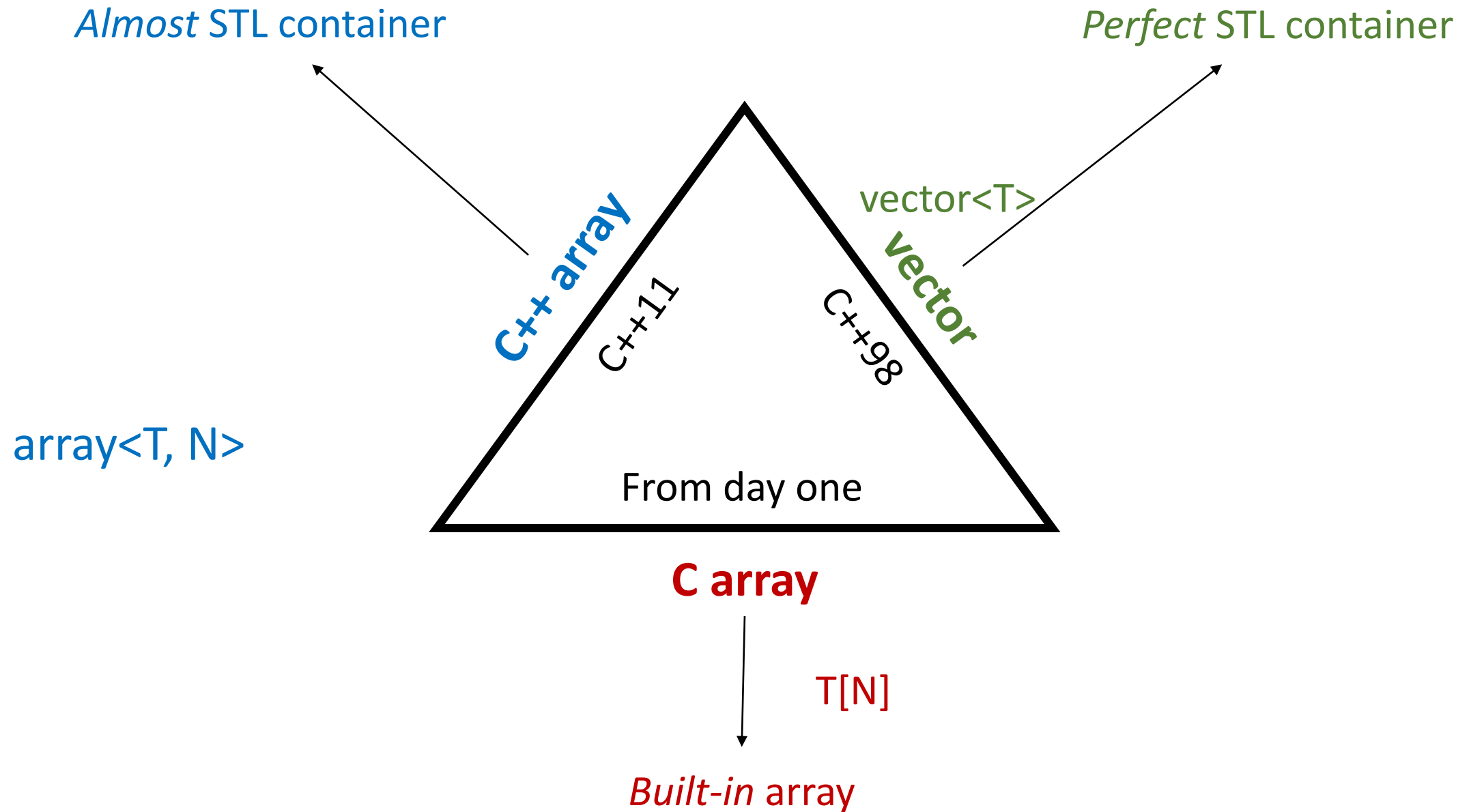
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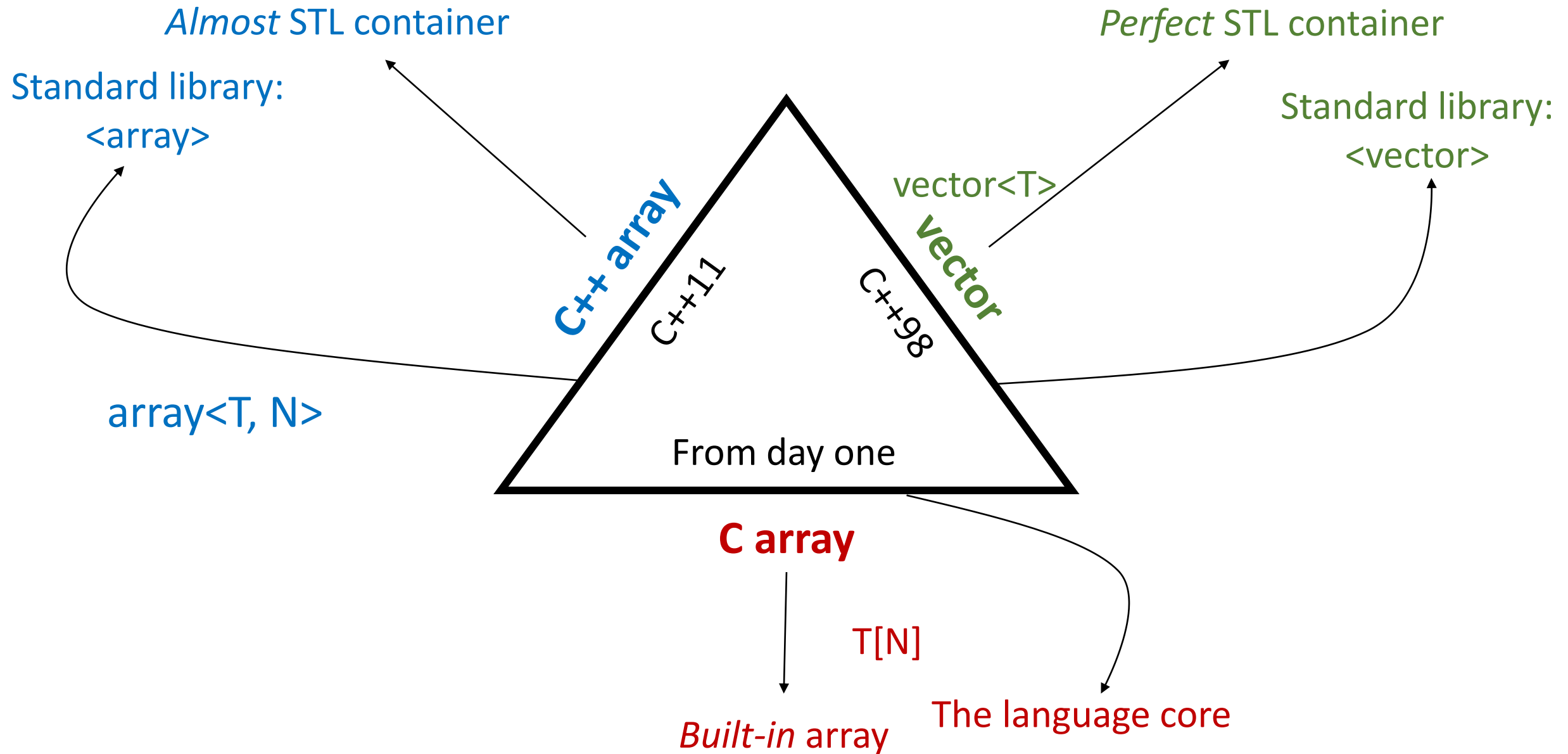
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Built-in array vs. Standard array vs. vector



Built-in array vs. Standard array vs. vector



Built-in array vs. Standard array vs. vector

Almost STL container

Perfect STL container

Standard library:

`<array>`

Compile-time
constant size

`array<T, N>`

C++ array

C++11

C++98

vector<T>
vector

Standard library:

`<vector>`

Run-time
variable size

From day one

Compile-time
constant size

C array

`T[N]`

Built-in array

The language core

Built-in array vs. Standard array vs. vector

Almost STL container

Perfect STL container

Standard library:

`<array>`

Compile-time
constant size

Size-aware

`array<T, N>`

C++ array

C++11

C++98

vector<T>
vector

Standard library:

`<vector>`

Size-aware

Run-time
variable size

From day one

Compile-time
constant size

Size unaware

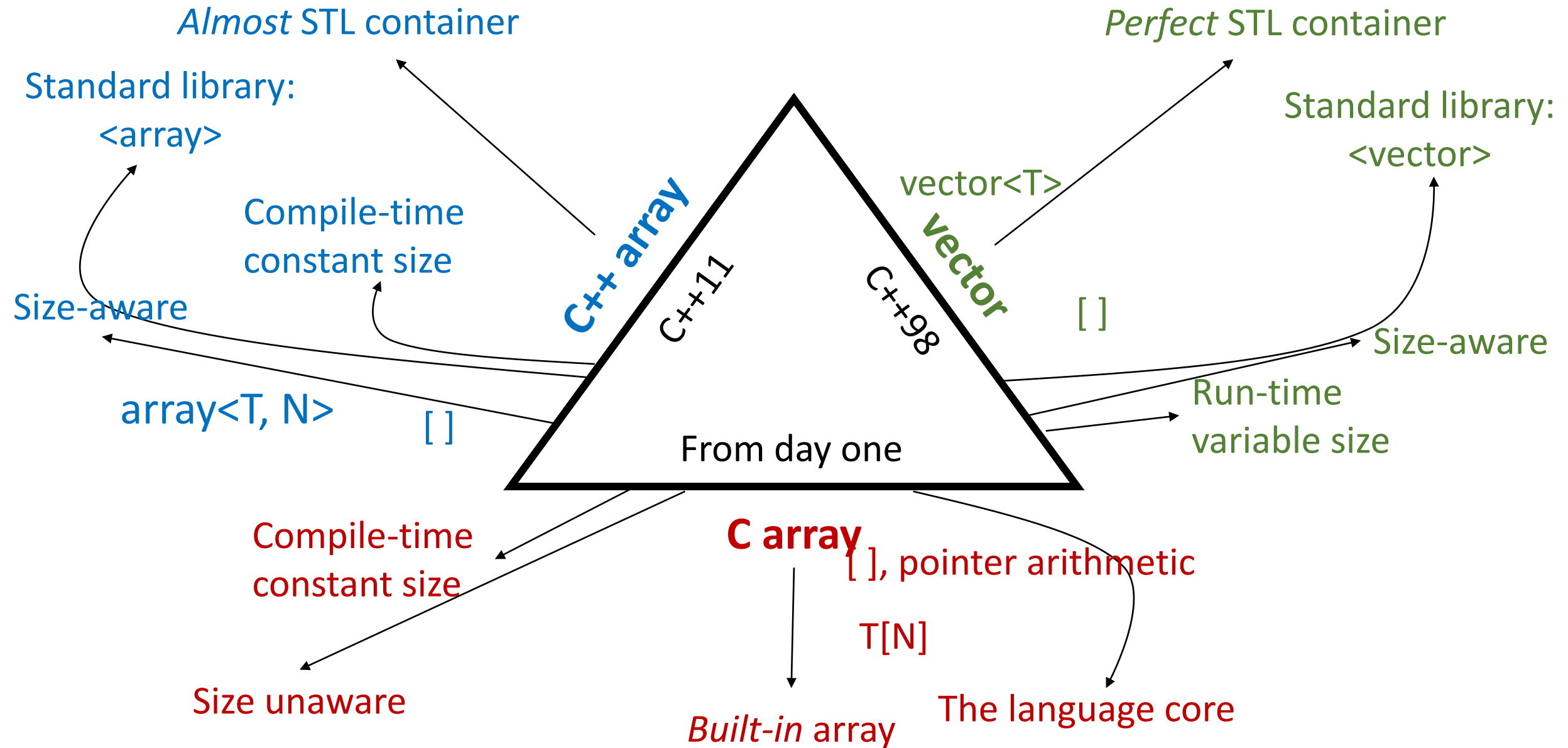
C array

`T[N]`

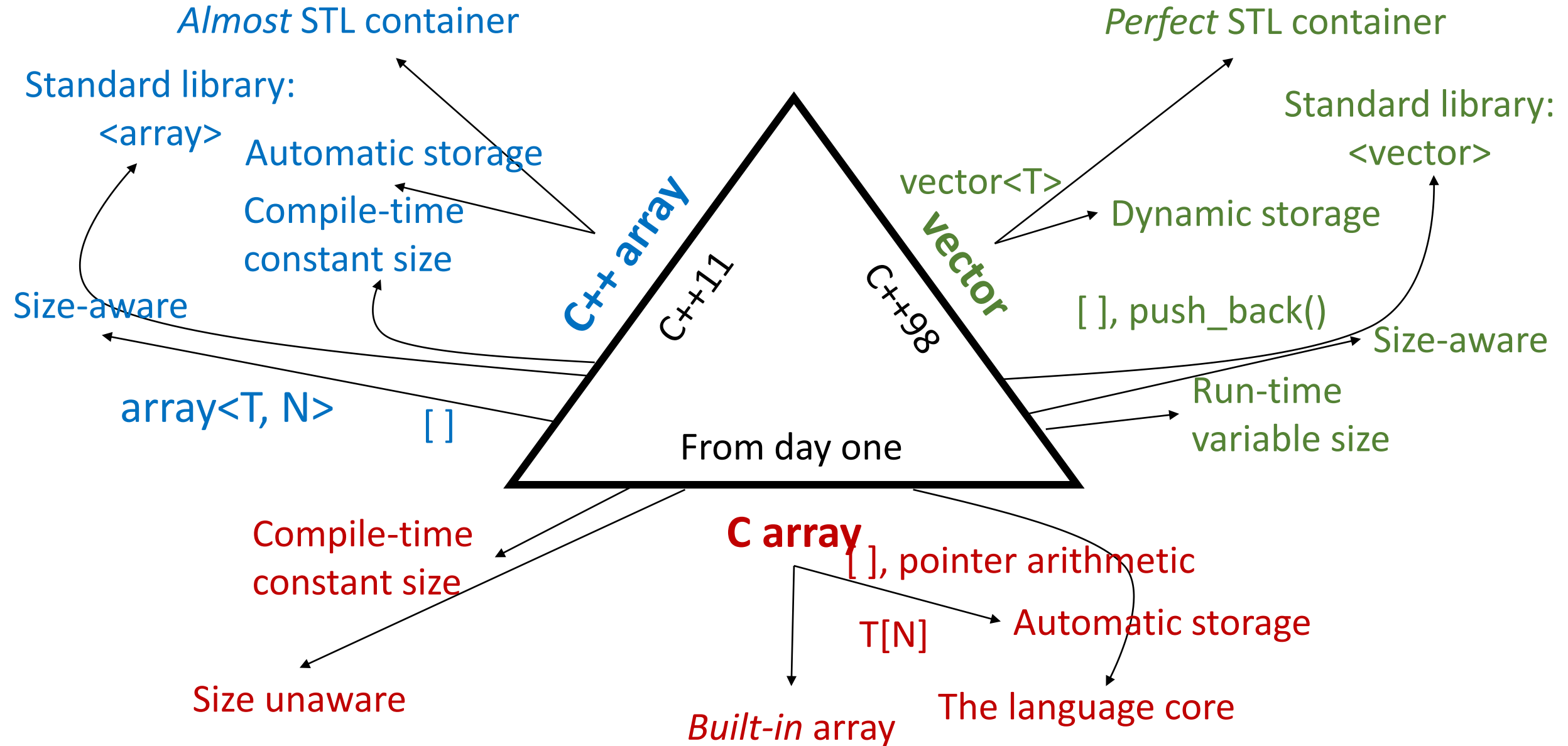
Built-in array

The language core

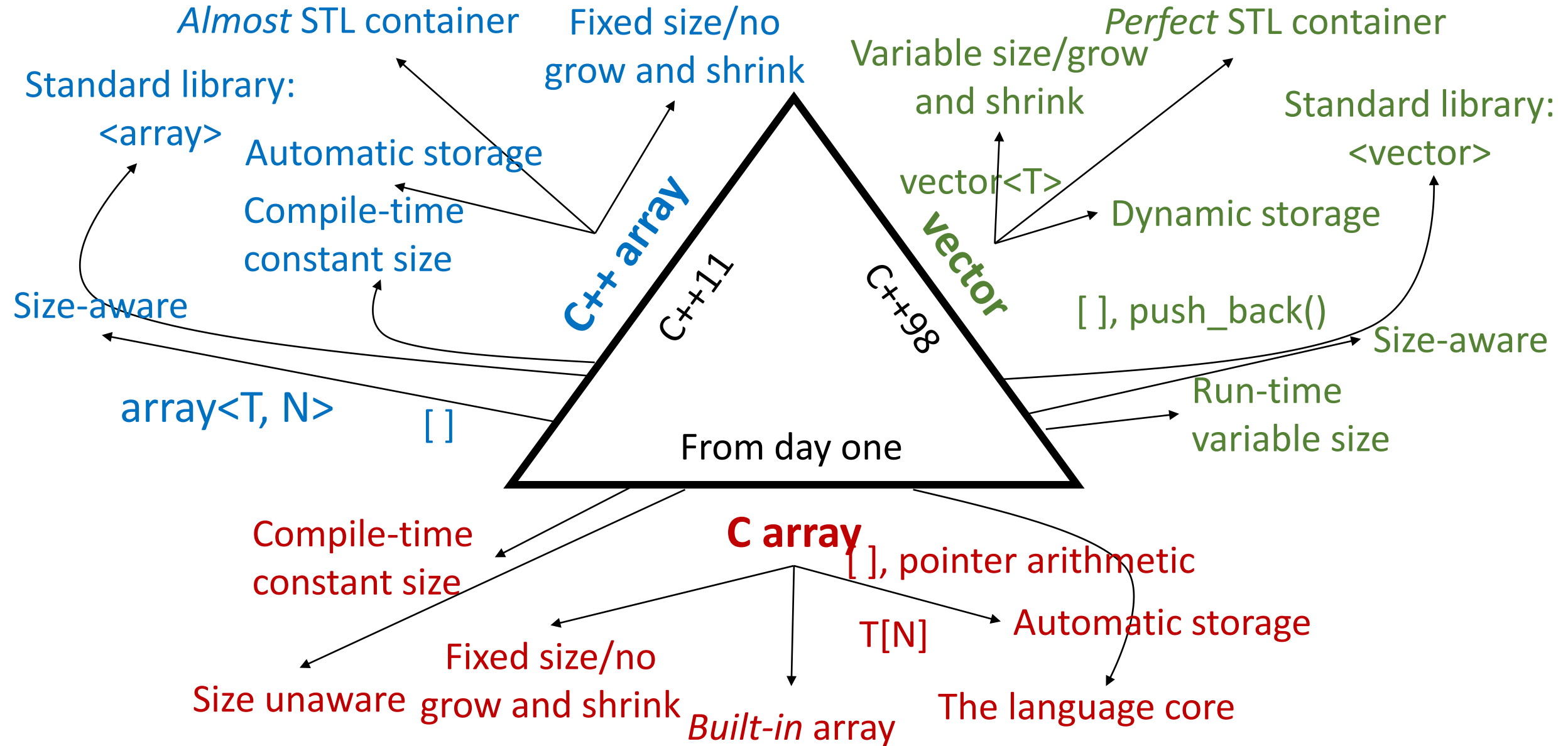
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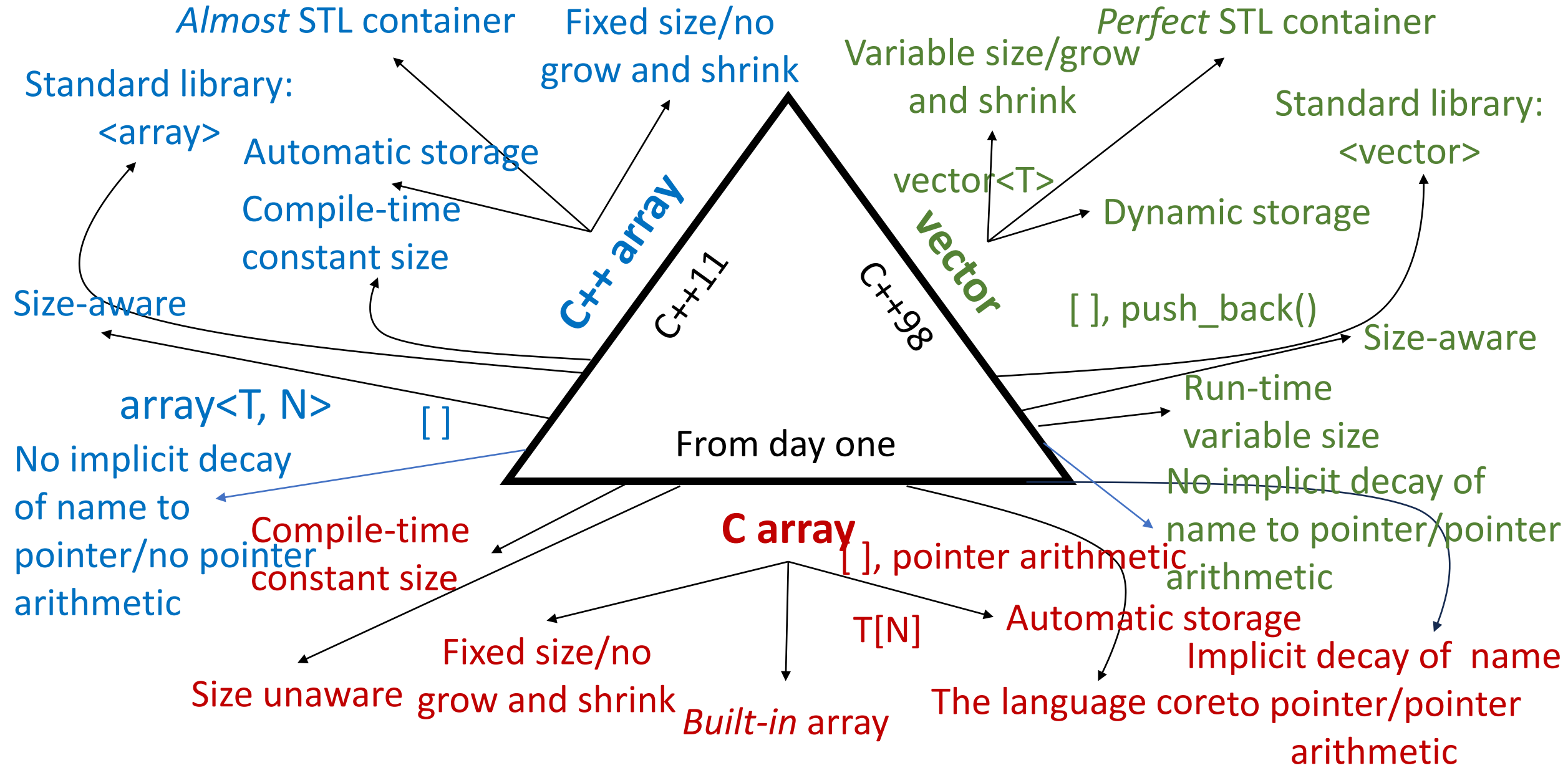
Built-in array vs. Standard array vs. vector



Built-in array vs. Standard array vs. vector



Built-in array vs. Standard array vs. vector



Rferences and further readings

- [Str00] Bjarne Stroustrup. The C++ Programming Language (Special Edition). Addison Wesley. Reading Mass. USA. February 2000.
- [Str08] Bjarne Stroustrup. Programming: Principles and Practice in C++, 1st edition, Addison Wesley, December 2008.
- [Str13] Bjarne Stroustrup. The C++ Programming Language. 4th edition. Addison Wesley. Reading Mass. USA. May 2013.
- [Str22] Bjarne Stroustrup. A Tour of C++, 3rd edition, Addison Wesley, September 2022.

Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

